Joonwoo Kwon

Ph.D. Student in Computer Science & Engineering, MSU
Physics-Informed Deep Learning, Computer Vision, Generative AI

Curriculum Vitae
Homepage: https://kwonjoon.info
Email: kwonjoon@msu.edu

Research Interests

Physics-Informed Deep Learning — integrating physical laws into deep generative and vision models to achieve interpretable and physically coherent 2D/3D generation, such as human motion generation.

Education

08/2025 – Present
East Lansing, MI
Ph.D. in Computer Science
Michigan State University (MSU)

03/2021 – 02/2023
Seoul, South Korea
M.S. in Applied Bioengineering
Seoul National University (SNU)

03/2015 – 02/2021
Suwon, South Korea
B.S. in Electronic and Electrical Engineering
SungKyunKwan University (SKKU) | (2016 – 2018) Korea Air Force for military service

Research Experience

08/2025 – Present
East Lansing, MI
MSU DominoAI Lab (Advisor: Dr. Zijun Cui)
Research Assistant | Physics-Informed Deep Learning & Generative Modeling

- Developed retargeting algorithm and datasets for physics-informed deep learning (e.g., human motion generation).

02/2023 – 12/2024
Seoul, South Korea
SNU Connectome Lab (Advisor: Dr. Jiook Cha)
Research Associate | Neuroscience & Generative Modeling

- Developed a new neural style transfer method **(C1)** for aesthetic-aware stylization.
- Designed an image-to-image translation model **(P1)** for cross-modal MRI synthesis.
- Proposed a novel generation task, dataset, and a multimodal framework **(C2)** for reconstructing video with music contextualized by human affect from brain signals.

02/2023 – 12/2024
Upton, NY (Remote)
Brookhaven National Lab (Advisor: Dr. Shinjae Yoo, Dr. Yuewei Lin)
Research Associate | Computer Vision & Multimodal Learning

- Developed a training-free approach for music style transfer **(P2)** by directly manipulating the self-attention features of pre-trained diffusion models.
- Designed viscosity-aware style optimization and brushstroke parameterization to emulate the physical and textural properties of oil painting and watercolor.
- Proposed a brain-to-text generation model and showed its versatility (e.g., composable brain decoding), inspired by how the brain perceives the visual world.

03/2022 – 06/2022
Seoul, South Korea
Samsung Advanced Institute of Technology (SAIT) (Research Capstone)
Student Researcher | Image-to-image translation, Semiconductor, and 3D Depth

- Led research on an image-to-image translation model utilizing U-NET and PatchGAN to synthesize 3D depth maps from SEM imaging.

Professional Experience

01/2025 – 05/2025
YongIn, South Korea
Hanwha Systems Co., Ltd. | Institute of Advanced Technologies (Defense & Space)
Research Scientist (Full-time) | *Military Satellite Imaging (SAR)*

- Developed image registration algorithms for SAR (Synthetic Aperture Radar) analysis.

10/2024 – 12/2024
Seoul, South Korea
Planningo Inc.
Research Engineer | Commercial Photography, Image Compositing

- Developed an image harmonization framework that resolves inconsistencies in lighting, textures, and color for commercial photography compositing.

Publications

[P2]. Stylus: Repurposing Stable Diffusion for Training-Free Music Style Transfer on Mel-Spectrograms
Wang, H.*, **Kwon, J.***, Kim, S.*, Seo, J., Yoo, S.†, Lin, Y.†, & Cha, J.†
(Under Review, 2026)

[P1]. Macro2Micro: Cross-modal Magnetic Resonance Imaging Synthesis Leveraging Multi-scale Brain Structures
Kim, S.*, **Kwon, J.***, Kwon, J.*, Bae S., Yoo, S.†, Lin, Y.†, & Cha, J.†
(Under Review, 2026)

* Equal contribution;
† corresponding author

[C2]. Revisiting Your Memory: Reconstruction of Affect-Contextualized Memory via EEG-guided Audiovisual Generation
Kwon, J.*, Wang, H.*, Lee, J.*, Kim, S.*, Yoo, S., Lin, Y.,† & Cha, J.†
ACM MM CogMAEC '25 (Oral)

[C1]. AesFA: An Aesthetic Feature-Aware Arbitrary Neural Style Transfer
Kwon, J.*, Kim, S.*, Yoo, S.†, Lin, Y.†, & Cha, J.†
AAAI 2024. Acceptance Rate: 23.75% (2342/12100).

Manuscripts in Preparation

[P5]. An Instance-Adaptive Photorealistic Style Optimization for Commercial Image Harmonization
Kim, S.*, **Kwon, J.***, Shin, J., Cha, J., & Kim, S. †

[P4]. Compositional Brain Decoding from Symbolic Representations in the Hierarchical Visual System
Kim, S.*, **Kwon, J.***, Wang, H., Kwon, J., Park, M. †, Yoo, S. †, Lin, Y. †, & Cha, J. †

[P3]. A Viscosity-guided Artistic Style Optimization via Brushstroke Parameterization
Kwon, J.*, Kim, S.*, Lee, S.*, Yoo, S., Lin, Y. †, & Cha, J.†

Selected Projects

10/2024 **The Recollection of Your Most Cherished Experience Utilizing AI and Neural Signals**
- Proposed a multimodal AI framework for synthesizing personalized video with music using generative AI and neural signals (EEG).

09/2023 – 08/2025 **Affect-Contextualized Perception Decoding with Cross-Species Multiscale Neuroscience Foundation Model**
- Developed a composable brain-to-text/image model using brain signals (fMRI)

09/2020 – 12/2020 **An Appreciation Aid Tool for the Visually Impaired via Synesthetic Perception**
- Developed an Arduino-based tool for the visually impaired that converted object colors and brightness into musical notes to enable synesthetic perception.

Honors and Awards

2024 **The Grand Prize,** **AI & Art Hackathon** ($1,000 USD), AI Art Research Center, SNU
2020 **Academic Excellence Scholarship for Outstanding Research** (25% tuition), SKKU
2020 **Corporate Partner Scholarship** (75% tuition), SKKU, ITECH Industrial Systems
2018 **The 2nd Winner for the 9th Engineering Competition for Local Impact**, SKKU
2018 **Korean Patent** (Applied; Public Telephone Booth for Sightseeing)

Invited Talks & Exhibition

10/2024 **ART DIFFUSION**, Tech to Art Platform (TAP) Prequel, SNU Museum of Art
09/2024 **Invited Talks:** A Composable Brain Decoding, Annual Meetings on Brain Decoding, SNU

Teaching & Mentoring

08/2018 – 12/2018 **Exchange Student Mentoring,** SKKU, (Electronic Circuits I; Introduction to Automatic Control)

Services

Reviewer **CVPR, ICIP, NeurIPS**

Skills

Communications **English** (Fluent; TOEFL 110; R30 L29 S24 W27), **Korean** (Native)
Programming Python, PyTorch, TensorFlow, MATLAB, C, R
Others **Hardware Languages** Verilog (intermediate), VHDL (intermediate)

Reference

Dr. Zijun Cui (Assistant Professor, MSU CSE | Doctoral Advisor) | email: cuizijun@msu.edu

Dr. Jiook Cha (Associate Professor, SNU Psychology | Academic Advisor) | email: connectome@snu.ac.kr

Dr. Shinjae Yoo (Chair, Artificial Intelligence Department, Brookhaven National Lab.)| email: sjyoo@bnl.gov

Dr. Yuewei Lin (Senior Computational Scientist, Brookhaven National Lab.) | email: ywlin@bnl.gov